Clustering and Initialization:

An Empirical Comparison of K-means and K-means++

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Abstract

K-means clustering is sensitive to the choice of its initial centroids because the clustering objective is non-convex. This report compares uniform random initialization with K-means++ (distance-squared seeding), while keeping the subsequent Lloyd iterations identical. Both algorithms are implemented from scratch using NumPy. We study a synthetic dataset of 600 well-separated two-dimensional observations and a reproducible subset of 5,000 MNIST handwritten-digit images represented by 784 features. For each dataset and initialization, we run 30 trials and analyze the distribution and variability of the final inertia. K-means++ reduces mean inertia by 75.75% on the synthetic dataset but only by approximately 0.10% on MNIST; the corresponding reductions in standard deviation are 68.49% and 35.31%. The experiments illustrate how careful initialization can substantially improve reliability, while its empirical benefit depends on the data geometry.

1 Introduction

Clustering is an unsupervised learning task that aims to organize observations into groups of similar data points without using class labels. One widely used algorithm is K-means, which partitions the data into k clusters by minimizing the within-cluster sum of squared Euclidean distances without using class labels. Although Lloyd's iterative procedure is simple, its solution depends on the initial centroid locations, and distinct initializations can converge to different local optima.

Arthur and Vassilvitskii proposed K-means++, a randomized seeding strategy that preferentially selects centers far from previously chosen centers [1].The basic idea is that a more spatially distributed set of initial centers can help avoid poor clustering solutions

This assignment investigates two questions: *(i)* whether K-means++ produces lower final inertia compared to randomly initializing the centers, and *(ii)* whether it reduces variability across random restarts. The comparison is performed on a well-separated synthetic dataset and on MNIST, a higher-dimensional real-world dataset. The latter allows us to test whether the same empirical result persists in a more complex feature space.

2 Algorithms

2.1 K-means inertia and Lloyd iterations

Let $X = \{x_1, \ldots, x_n\} \subset \mathbb{R}^d$ be a dataset, and let $\mu_1, \ldots, \mu_k \in \mathbb{R}^d$ denote the cluster centroids. For assignments $c_i \in \{1, \ldots, k\}$, the K-means inertia is

$$J(c, \mu) = \sum_{i=1}^{n} \|x_i - \mu_{c_i}\|_2^2 . \tag{1}$$

Lloyd's algorithm alternates two operations. In the *assignment step*, every observation is assigned to the nearest current centroid:

$$c_i \leftarrow \arg\min_{j \in \{1,\ldots,k\}} \|x_i - \mu_j\|_2^2. \tag{2}$$

In the update step, each nonempty cluster centroid is updated to the mean of its assigned points.

$$\mu_j \leftarrow \frac{1}{|C_j|} \sum_{i \in C_j} x_i, \qquad C_j = \{i : c_i = j\}. \tag{3}$$

These two steps do not increase the inertia when carried out exactly. However, they do not guarantee convergence to a globally optimal clustering because the optimization problem is non-convex.

2.2 Initialization strategies

Uniform random initialization. Select k *distinct* observations uniformly without replacement and use them as initial centroids. This baseline is inexpensive but may place multiple initial centers in the same region, leaving other clusters initially unrepresented.

K-means++ initialization. Choose the first center uniformly at random. Given the centers already selected, define the squared distance of each point x to its nearest selected center as

$$D(x)^2 = \min_{\mu \in C} \|x - \mu\|_2^2, \tag{4}$$

where C is the set of selected centers. Sample each subsequent center with probability

$$\Pr(x \text{ is selected}) = \frac{D(x)^2}{\sum_{z \in X} D(z)^2}. \tag{5}$$

This process repeats until k centers are chosen. If the total weight is zero, all points have zero distance to at least one selected center. In this case, the implementation selects uniformly among the indices not yet chosen. Both initialization strategies then use exactly the same Lloyd iterations. Thus, differences in the measured outcomes can be attributed to the initialization procedure, subject to sampling variability.

2.3 Implementation details

Both methods are coded from scratch with NumPy; no library K-means estimator is used. Squared distances between a point x and a center μ are evaluated with the algebraic identity

$$\|x - \mu\|_2^2 = \|x\|_2^2 + \|\mu\|_2^2 - 2x^{\mathsf{T}}\mu, \tag{6}$$

which allows vectorized matrix multiplication without allocating a large $n \times k \times d$ array.

The procedure uses at most 100 iterations and stops earlier when the old and updated centroids satisfy NumPy's default `allclose` criterion. If a cluster is empty, its centroid remains at its previous location. Final assignments and inertia are computed from the last centroids. The same update logic is shared by both initialization methods.

3 Datasets and Experimental Protocol

3.1 Synthetic dataset

The synthetic dataset contains $n = 600$ observations in $\mathbb{R}^2$, generated with `sklearn.datasets.make_blobs`.[3] Four clusters are centered at $(-8, -8)$, $(-8, 8)$, $(8, -8)$ and $(8, 8)$, with standard deviation 1.3. The dataset-generation seed is 123, and we set $k = 4$. The large distances between the four centers produce clear cluster structure, making incorrect seeding especially visible.

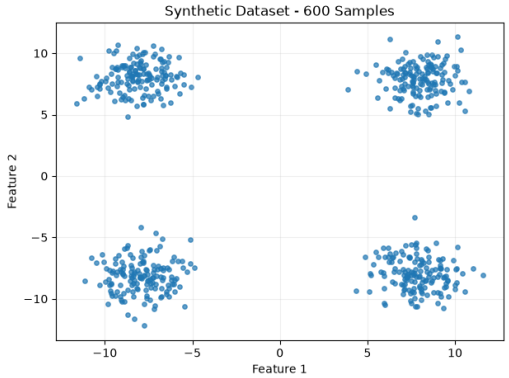


Figure 1: The two-dimensional synthetic dataset used for the clustering experiments (600 observations). The four groups are well separated.

3.2 MNIST dataset

MNIST consists of handwritten-digit images from ten classes[2]. We use a reproducible random sample of $n = 5{,}000$ images from its 60,000-image training partition, sampled without replacement using seed 42. Each 28×28 grayscale image is flattened to a vector in $\mathbb{R}^{784}$, and pixel values are divided by 255 to lie in $[0, 1]$. We choose $k = 10$, matching the number of digit classes. The digit labels are used only to display example images and *never* enter the clustering algorithm.

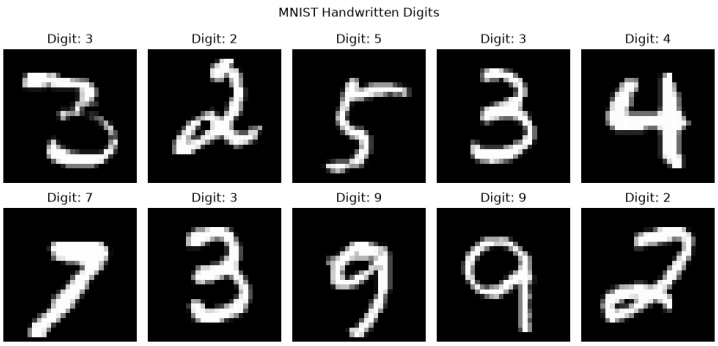


Figure 2: Examples from the sampled MNIST images. Digit labels are shown for illustration only; clustering is unsupervised.

3.3 Repetitions and evaluation metrics

For each dataset, we perform 30 independent runs of each initialization strategy, using random seeds 0 through 29. Dataset, cluster count, iteration limit, and Lloyd updates are fixed within each comparison. Each run outputs the final inertia in Equation (1). We report its *mean*, *standard deviation*, *median*, *minimum*, and *maximum*, as well as boxplots summarizing its distribution.

For a metric m for which lower values are preferable, we quantify the relative decrease as

$$\text{Relative decrease}(m) = 100\,\frac{m_{\text{random}} - m_{++}}{m_{\text{random}}}\,\%. \tag{7}$$

The standard deviation summarizes run-to-run variability *in achieved inertia*, rather than similarity between the actual cluster memberships. All standard deviations below use the population formula (division by the number of runs).

4 Results

4.1 Synthetic dataset

Table 1 shows a remarkable difference in performance. The mean inertia falls from 10,830.32 under random initialization to 2,625.91 with K-means++; the decrease is 75.75%. Standard deviation drops from 10,650.61 to 3,356.11 (68.49%). Both strategies occasionally reach the same best observed inertia, approximately 2,002.70, which is also the median for both methods. K-means Random doesn't always produce worse results. In fact, on the synthetic dataset, Random and K-means++ have the same median. However, Random has some runs with very high inertia, which raises the average. K-means++ reduces the frequency of these bad solutions.

Table 1: Final inertia on the synthetic dataset, across 30 runs.

Initialization	Mean	Std. dev.	Median	Minimum
Random	10,830.32	10,650.61	2,002.70	2,002.70
K-means++	2,625.91	3,356.11	2,002.70	2,002.70

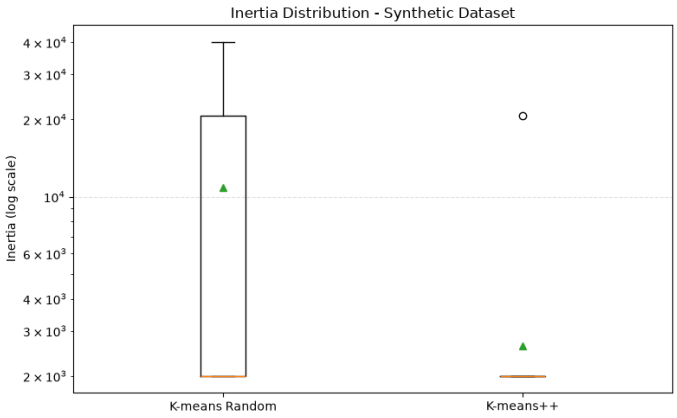


Figure 3: Final inertia distributions for the synthetic dataset. The logarithmic vertical scale accommodates the large spread of random-initialization outcomes.

We defined a near-best run as one where its final inertia doesn't exceed the minimum inertia observed between the two methods by more than 5%. The corresponding threshold

is approximately 2,102.83. On the synthetic dataset, 17 out of 30 random-initialization runs (56.7%) and 29 out of 30 K-means++ runs (96.7%) are classified as near-best. This shows that K-means++ more often reaches good solutions. However, the 5% threshold is an empirical criterion chosen after the experiment and doesn't guarantee that we've found the global minimum.

4.2 MNIST dataset

The MNIST results in Table 2 show a smaller but consistent descriptive difference. The mean inertia is 196,865.36 with random initialization and 196,672.14 with K-means++, equivalent to a reduction of approximately 0.10%. The standard deviation decreases from 646.09 to 417.94 (35.31%), indicating less variability in the attained inertia values. The best observed inertia values are close, at 196,153.53 and 196,135.53, respectively.

Table 2: Final inertia on the 5,000-image MNIST subset, across 30 runs.

Initialization	Mean	Std. dev.	Median	Minimum	Maximum
Random	196,865.36	646.09	196,723.45	196,153.53	198,598.44
K-means++	196,672.14	417.94	196,632.73	196,135.53	197,526.03

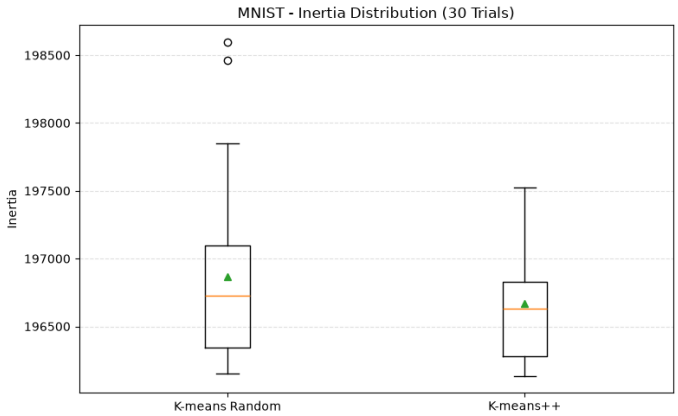


Figure 4: Distribution of final inertia for MNIST across 30 runs. A linear scale is used to make relatively small numerical differences visible.

4.3 Comparison across datasets

Table 3 compares the *relative* improvement from K-means++ within each dataset. Both datasets exhibit lower mean inertia and lower standard deviation for K-means++ in these experiments, but the magnitude of the mean improvement is markedly different.

Table 3: Relative reductions from random initialization to K-means++.

Dataset	Mean inertia reduction	Std. deviation reduction
Synthetic	75.75%	68.49%
MNIST	0.10%	35.31%

It would be misleading to compare raw inertia directly between these datasets, since they differ in sample count, feature dimension, feature scaling, and chosen number of clusters. Within-

dataset relative comparisons instead isolate the empirical effect of initialization under each experimental setting.

5 Discussion

The synthetic dataset is constructed with four very well-separated groups. Uniformly selecting all starting centers from the observed points can choose multiple points from one group and none from another. Lloyd's method can then converge to a suboptimal partition. K-means++ mitigates this risk by favoring observations that lie far from existing centers. The sharp reduction in high-inertia runs is consistent with this mechanism.

On the synthetic dataset, K-means++ achieved a much higher average improvement because the groups are clearly separated. On MNIST, on the other hand, the data has a more complex structure and the classes don't necessarily correspond to well-separated spherical clusters in pixel space. Also, K-means minimizes inertia, it doesn't try to recognize the digits. However, we can't prove that the different cluster separation is the only reason, because the two datasets also differ in dimensionality, number of points, and number of clusters.

Our project has some limitations. We only ran 30 trials, so we need to interpret the small differences observed on MNIST with caution. In addition, we used a fixed value of (k) per dataset and a numerical convergence criterion that could affect the results. Finally, we measured the stability of the inertia, not that of the cluster assignments. As possible extensions, we could measure the number of iterations needed to reach convergence and evaluate K-means and K-means++ on synthetic datasets with progressively more overlapping clusters.

6 Conclusion

In 30-run experiments on two datasets, K-means++ achieves lower mean final inertia and lower run-to-run variability than uniform random initialization. The impact is particularly strong for the separated synthetic clusters, where the mean inertia decreases by 75.75%, but substantially smaller on the sampled MNIST data, where it decreases by about 0.10%. The standard deviation decreases in both cases, suggesting more consistent inertia values. These findings support the practical importance of initialization in non-convex clustering, while also illustrating that the magnitude of the benefit depends on the dataset.

References

[1] D. Arthur and S. Vassilvitskii. k-means++: The Advantages of Careful Seeding. In *Proceedings of the Eighteenth Annual ACM-SIAM Symposium on Discrete Algorithms (SODA)*, 2007.

[2] Y. LeCun, L. Bottou, Y. Bengio, and P. Haffner. Gradient-Based Learning Applied to Document Recognition. *Proceedings of the IEEE*, 86(11):2278–2324, 1998.

[3] F. Pedregosa et al. Scikit-learn: Machine Learning in Python. *Journal of Machine Learning Research*, 12:2825–2830, 2011.

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